

Solution

1. (Binary stones). Consider the following game: two players may take from a heap any number of stones that is a power of 2 (including $1 = 2^0$). The player who has taken the last stone wins. For which number of stones does the first player win? How should he play? (You should not only specify the strategy but also prove that it works.)

Solution.

Let n be the number of stones. Allowed moves are to remove $1, 2, 4, 8, \dots, 2^k$ stones. We seek the losing positions:

Small cases.

n	0	1	2	3	4	5	6	7	8	9
outcome	lose	win	win	lose	win	win	lose	win	win	lose

The losing positions among these are $0, 3, 6, 9, \dots$

Claim. The losing positions are exactly the multiples of 3. Equivalently, the first player wins if and only if $n \not\equiv 0 \pmod{3}$. For example, 1, 2 or 4 and etc.

Proof. Powers of 2 modulo 3 alternate:

$$2^0 \equiv 1 \pmod{3} \quad 2^1 \equiv 2 \pmod{3}$$

$$2^2 \equiv 1 \pmod{3} \quad 2^3 \equiv 2 \pmod{3}$$

In general $2^k \equiv (-1)^k \pmod{3}$, so every allowed move removes either 1 or 2 modulo 3, never 0 modulo 3.

If $n \equiv 0 \pmod{3}$, then for any move $n \mapsto n - 2^k$ we have $n - 2^k \not\equiv 0 \pmod{3}$. If $n \not\equiv 0 \pmod{3}$, there is a move to a multiple of 3:

1. $n \equiv 1 \pmod{3}$, remove $1 = 2^0$
2. $n \equiv 2 \pmod{3}$, remove $2 = 2^1$

Thus from a multiple of 3 every move goes to a non-multiple of 3, and from a non-multiple of 3 there is a move to a multiple of 3. The terminal position $n = 0$ is a multiple of 3 and is losing for the player to move. Hence multiples of 3. Therefore the first player wins if and only if $n \not\equiv 0 \pmod{3}$.

Optimal strategy. If $n \equiv 1 \pmod{3}$, remove 1 stone; if $n \equiv 2 \pmod{3}$, remove 2 stones. Thereafter always leave a heap size divisible by 3. Since the opponent can only subtract 1 or 2 modulo 3 from the heap, you can always restore divisibility by 3 on your turn. Eventually the opponent faces $n = 0$ and loses.

2. (Moving the knight).

We number the board by coordinates

$$a1 = (1, 1), \quad h8 = (8, 8).$$

The knight starts at

$$h8 = (8, 8).$$

A legal move is one of

$$(x, y) \mapsto (x + 1, y - 2),$$

$$(x, y) \mapsto (x - 1, y - 2),$$

$$(x, y) \mapsto (x - 2, y + 1),$$

$$(x, y) \mapsto (x - 2, y - 1)$$

provided the new square is still on the board. winning if the player to move from it can move to a losing position.

Losing positions on the 8×8 board

In coordinates, these are the 2×2 blocks

$$\{1, 2\} \times \{1, 2\},$$

$$\{1, 2\} \times \{5, 6\},$$

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together with the special square

$$(8, 8).$$

We can see that losing positions satisfy

$$x, y \equiv 1 \text{ or } 2 \pmod{4}$$

On the 8×8 board, the losing positions can be displayed as follows:

	1	2	3	4	5	6	7	8
8	.	.	.	.	.	.	.	<i>L</i>
7	.	.	.	.	.	.	.	.
6	<i>L</i>	<i>L</i>	.	.	<i>L</i>	<i>L</i>	.	.
5	<i>L</i>	<i>L</i>	.	.	<i>L</i>	<i>L</i>	.	.
4	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.
2	<i>L</i>	<i>L</i>	.	.	<i>L</i>	<i>L</i>	.	.
1	<i>L</i>	<i>L</i>	.	.	<i>L</i>	<i>L</i>	.	.

Why these positions are losing

Let

$$R = \{(x, y) : x, y \equiv 1 \text{ or } 2 \pmod{4}\}.$$

These positions form repeated 2×2 blocks. A knight move changes one coordinate by 1 and the other coordinate by 2. Therefore a move from a position in R always leaves R . So no position in R has a move to another position in R . Conversely, by backward induction, every legal move out of one of these blocks leads to a position from which the next player can move back into a smaller R -position. Hence the positions in R are losing. Now check the starting square:

$$h8 = (8, 8)$$

Its only legal moves are

$$(8, 8) \mapsto (7, 6)$$

and

$$(8, 8) \mapsto (6, 7)$$

Both of these positions are winning, because both can move to

$$(5, 5) = e5,$$

which is losing:

$$(7, 6) \mapsto (5, 5),$$

$$(6, 7) \mapsto (5, 5)$$

Therefore every legal move from $h8$ goes to a winning position. Hence

$$h8$$

is losing.

Thus the first player loses and the second player wins.

General $n \times n$ board

Now suppose the board is $n \times n$, and the knight starts at the top-right corner

$$(n, n).$$

The losing block pattern repeats modulo 4. Namely, positions with

$$x, y \equiv 1 \text{ or } 2 \pmod{4}$$

are losing positions.

Now consider the starting square (n, n) .

Case 1: $n \equiv 1 \pmod{4}$

Then

$$(n, n) \equiv (1, 1) \pmod{4}.$$

So (n, n) lies in a losing block. Therefore the second player wins.

Case 2: $n \equiv 2 \pmod{4}$

Then

$$(n, n) \equiv (2, 2) \pmod{4}.$$

Again, (n, n) lies in a losing block. Therefore the second player wins.

Case 3: $n \equiv 3 \pmod{4}$

Then the first player can move to a losing block:

$$(n, n) \mapsto (n-1, n-2).$$

Modulo 4,

$$(3, 3) \mapsto (2, 1).$$

Since $(2, 1)$ is a losing-type residue, the first player moves to a losing position. Therefore the first player wins.

Case 4: $n \equiv 0 \pmod{4}$

The only legal moves from (n, n) are

$$(n, n) \mapsto (n-1, n-2),$$

$$(n, n) \mapsto (n-2, n-1).$$

But both resulting positions are winning, because each can move to

$$(n-3, n-3),$$

and

$$(n-3, n-3) \equiv (1, 1) \pmod{4}.$$

So $(n-3, n-3)$ is losing.

Therefore (n, n) is losing, and the second player wins.

Final general result

Hence, on an $n \times n$ board starting from the top-right corner,

the first player wins if and only if $n \equiv 3 \pmod{4}$.

For the usual chessboard,

$$8 \equiv 0 \pmod{4},$$

so the first player loses and the second player wins.